

Lesson 23 — Using a Variable in a Decision

Matches LEGO Teacher Portal lesson: "Feed and Play".

Objective (I can...): I can use a variable inside an if/then so my program reacts to stored data. **CS concept:** Variables combined with conditionals · **Builds on:** Lesson 22 (counting) and Lesson 18 (if/then/else).

AI Connection (2 min): AI acts on stored data: IF the count of correct guesses is high enough, THEN it's "trained." *First-principles question:* "What number would tell a robot it's ready?"

Materials: LEGO CS & AI kit + device per pair, trigger object/button. **Model reuse:** Variables model from Lesson 21 — code only.

60-minute flow - Warm-up (8): "When does it change?" — coach counts claps with a rule ("at 5, everyone cheers"). - **Concept + demo (10):** Demo: count presses; if score reaches a target, then celebrate (light + sound); else keep counting. - **Build/Code (30):** Pairs code a counter plus an if/then that triggers a special action when the variable hits a chosen value. *Swap at 15 min.* - **Challenge (7):** Change the target value or add an else branch for "not yet." - **Share + pack-up (5):** Pairs demo reaching the target and the model reacting. Model stays built.

Pair roles: Driver codes the if/then, Navigator owns the counter and counts to the target. Swap at 15 min. (*Trio rotates.*)

Differentiation: *Support* — provide the counter; students add the if/then as framed pseudocode. *Extension / fast-finisher* — two targets with different reactions. *Reteach* — set the target to 2 so it triggers fast, then raise it.

Exit check: Pair's model changes behavior when the variable reaches the target. **At-home (optional):** Write a rule like "when my reading minutes reach 20, I earn a break." ~10 min.
